Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A long straight cylinder of radius R carries a current I which is uniformly distributed across its cross-section. Find the magnetic field at a point at a distance r from the axis of the cylinder in cases (a) $r > R$ and (b) $r < R$.

(A) $B = \frac{\mu_0 I}{2\pi r}$ and $B = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$

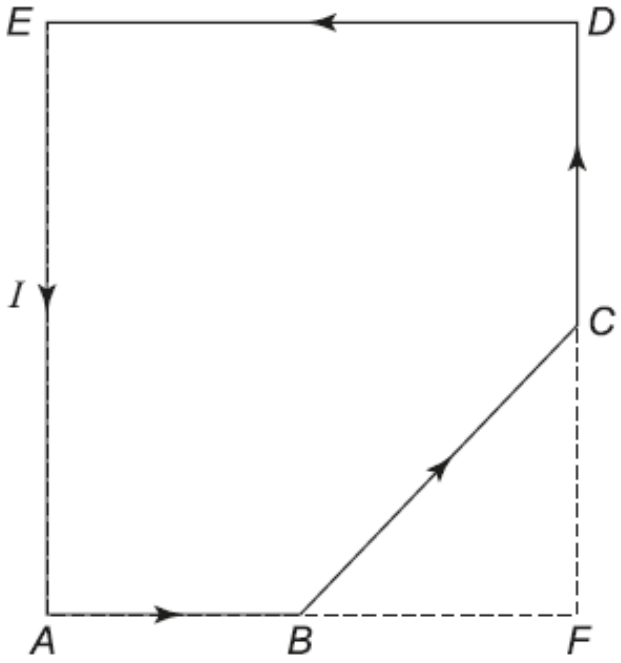
(B) $B = \frac{\mu_0 I}{4\pi r}$ and $B = \left(\frac{\mu_0 I}{8\pi R^2}\right) r$

(C) $B = \frac{\mu_0 I}{3\pi r}$ and $B = \left(\frac{\mu_0 I}{5\pi R^2}\right) r$

(D) $B = \frac{\mu_0 I}{8\pi r}$ and $B = \left(\frac{\mu_0 I}{4\pi R^2}\right) r$

Q.2

Figure shows a wire loop $ABCDEA$ carrying a current I as shown. Given $AE = ED = a$ and $AB = CD = a/2$. Find the magnitude and direction of the magnetic field at point F where $BF = CF = a/2$



(A) $\frac{\mu_0 I}{\pi a}\left(1 - \frac{4}{2\sqrt{2}}\right)$

(B) $\frac{\mu_0 I}{\pi a}\left(3 - \frac{4}{2\sqrt{2}}\right)$

(C) $\frac{\mu_0 I}{\pi a}\left(1 - \frac{1}{2\sqrt{2}}\right)$

(D) $\frac{\mu_0 I}{\pi a}\left(3 - \frac{1}{2\sqrt{2}}\right)$

@IITJEE_Advanced

Q.3

Two wires A and B have the same length L and carry equal currents I. Wire A is bent into a circle and wire B is bent into a square. What is the ratio of magnetic field produced by wire A and wire B ?

(A) 1.18

(B) 1.15

(C) 1.16

(D) 1.2

Q.4

Charge q is uniformly spread on a thin ring of radius R. The ring rotates about its axis with a uniform frequency fHz. The magnitude of magnetic induction at the centre of the ring is

(A) $\frac{\mu_0 \mathrm{qf}}{2\mathrm{R}}$

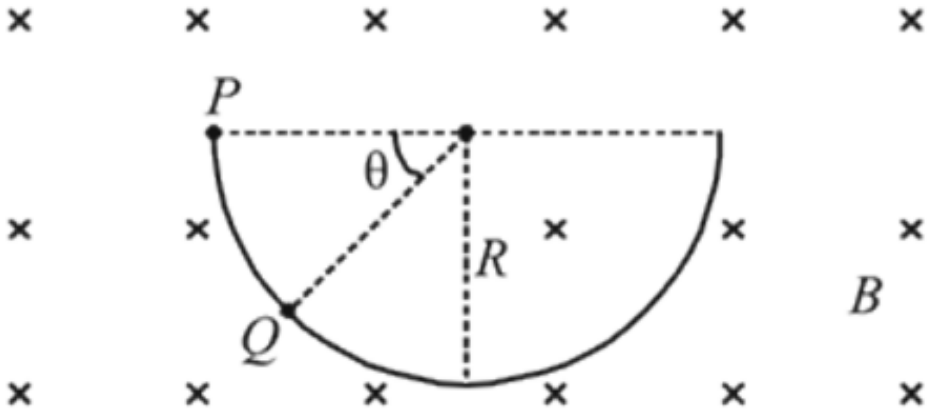
(B) $\frac{\mu_0 q}{2fR}$

(C) $\frac{\mu_0 q}{2\pi fR}$

(D) $\frac{\mu_0 qf}{2\pi R}$

Q.5

A charged sphere of mass m and charge $-q$ starts sliding along the surface of a smooth hemispherical bowl, at position P. The region has a transverse uniform magnetic field B. Normal force by the surface of bowl on the sphere at position Q is



(A) $mg\sin\theta + qB\sqrt{2gR\sin\theta}$

(B) $3mg\sin\theta + qB\sqrt{2gR\sin\theta}$

(C) $mg\sin\theta - qB\sqrt{2gR\sin\theta}$

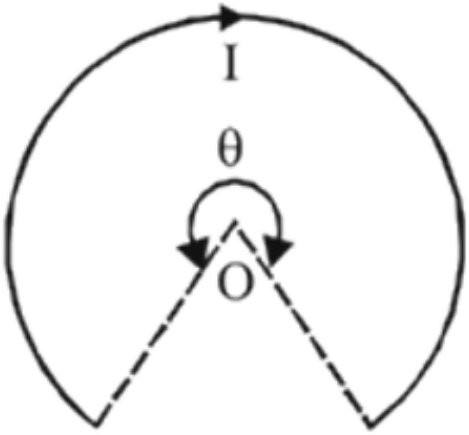
(D) $3mg\sin\theta - qB\sqrt{2gR\sin\theta}$

Q.6

A current of I ampere flows in a wire forming a circular arc of radius r metres subtending an angle θ at the

@IITJEE_Advanced

centre as shown. The magnetic field at the centre O in tesla is



(A) $\frac{\mu_0 I\theta}{4\pi r}$

(B) $\frac{\mu_0 I\theta}{2\pi r}$

(C) $\frac{\mu_0 I\theta}{2r}$

(D) $\frac{\mu_0 I\theta}{4r}$

Q.7

The magnetic field due to a current carrying circular loop of radius $3\ \mathrm{cm}$ at a point on the axis at a distance of $4\ \mathrm{cm}$ from the centre is $54\mu\mathrm{T}$. What will be its value at the centre of loop?

(A) $125\mu\mathrm{T}$

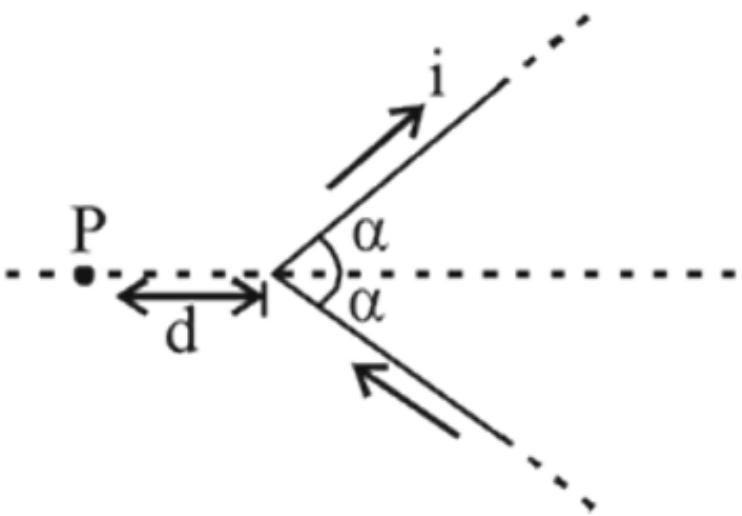
(B) $150\mu\mathrm{T}$

(C) $250\mu\mathrm{T}$

(D) $75\mu\mathrm{T}$

Q.8

If the magnetic field at P can be written as $\mathrm{K}\tan\left(\frac{\alpha}{2}\right)$, then K is



(A) $\frac{\mu_0 I}{4\pi d}$

@IITJEE_Advanced

(B) $\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}$

(C) $\frac{\mu_0 I}{\pi d}$

(D) $\frac{2\mu_0 \mathrm{I}}{\pi \mathrm{d}}$

Q.9

A wire carries a current. Maintaining the same current it is bent first to form a circular plane coil of one turn which produces a magnetic field B at the centre of the coil. The same length is now bent more sharply to give a double loop of smaller radius. The magnetic field at the centre of the double loop caused by the same current is

(A) $4B$

(B) $B/4$

(C) $B/2$

(D) $2B$

Q.10

Two long parallel wires P and Q are both perpendicular to the plane of the paper with distance of $5\ \mathrm{m}$ between them. If P and Q carry currents of 2.5 amp and 5 amp respectively in the same direction, then the magnetic field at a point half-way between the wires is

(A) $\frac{3\mu_0}{2\pi}$

(B) $\frac{\mu_0}{\pi}$

(C) $\frac{\sqrt{3}\mu_0}{2\pi}$

(D) $\frac{\mu_0}{2\pi}$

Q.11

Consider two thin identical conducting wires covered with very thin insulating material. One of the wires is bent into a loop and produces magnetic field B_1, at its centre when a current I passes through it. The second wire is bent into a coil with three identical loops adjacent to each other produces magnetic field B_2 at the centre of the loops when current $\frac{\mathrm{I}}{3}$ passes through it. The ratio $\mathrm{B}_1 : \mathrm{B}_2$ is :

(A) $1 : 1$

(B) $1 : 3$

(C) $1 : 9$

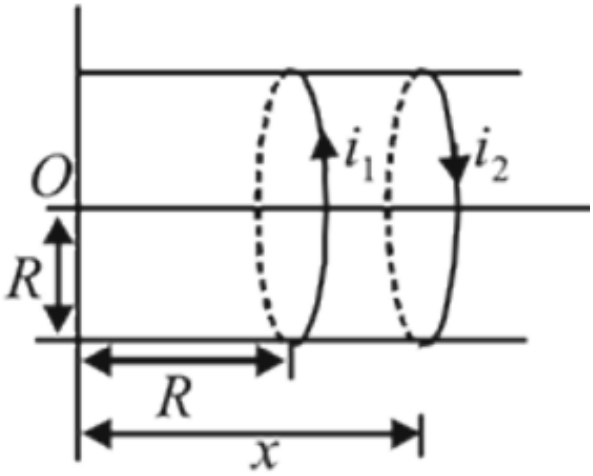
(D) $9 : 1$

Q.12

Two wires are wrapped over a wooden cylinder to form two coaxial loops carrying current i_1 and i_2. If $i_2 =$

@IITJEE_Advanced

$8i_1$ the value of x for $B = 0$ at the origin O is :



(A) $\sqrt{(\sqrt{7}-1)R}$

(B) $\sqrt{5}R$

(C) $\sqrt{3}R$

(D) $\sqrt{7}R$

Q.13

The current density $\vec{J}$ inside a long, solid, cylindrical wire of radius $a = 12\ \text{mm}$ is in the direction of the central axis, and its magnitude varies linearly with radial distance r from the axis according to $J = \frac{J_0 r}{a}$, where $J_0 = \frac{10^5}{4\pi}\text{A/m}^2$.
Find the magnitude of the magnetic field at $r = \frac{a}{2}$ in μT.

(A) $10\mu\text{T}$

(B) $4\mu\text{T}$

(C) $5\mu\text{T}$

(D) $3\mu\text{T}$

Q.14

A small current element of length $d\ell$ and carrying current is placed at $(1, 1, 0)$ and is carrying current in ' $+z$ ' direction. If magnetic field at origin be $\vec{B}_1$ and at $\text{point}(2, 2, 0)$ be $\vec{B}_2$ then:

(A) $\vec{B}_1 = \vec{B}_2$

(B) $\left|\vec{B}_1\right| = \left|2\vec{B}_2\right|$

(C) $\vec{B}_1 = -\vec{B}_2$

(D) $\vec{B}_1 = -2\vec{B}_2$

@IITJEE_Advanced

Q.15

A point charge of $0.1\mathrm{C}$ is placed on the circumference of a non-conducting ring of radius $1\ \mathrm{m}$ which is rotating with a constant angular acceleration of $1\mathrm{rad/sec}^2$. If ring starts its motion at $t = 0$, the magnetic field at the centre of the ring at $t = 10\mathrm{sec}$, is

(A) $10^{-6}\ \mathrm{T}$

(B) $10^{-7}\ \mathrm{T}$

(C) $10^{-8}\ \mathrm{T}$

(D) $10^{7}\ \mathrm{T}$

Q.16

A galvanometer of 50 ohm resistance has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one per division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of

(A) 2450Ω in series

(B) 2500Ω in series.

(C) 245Ω in series.

(D) 2550Ω in series.

Q.17

A beam of electrons is moving with constant velocity in a region having simultaneous perpendicular electric and magnetic fields of strength $20\mathrm{Vm}^{-1}$ and $0.5\ \mathrm{T}$ respectively at right angles to the direction of motion of the electrons. Then the velocity of electrons must be

(A) $8\ \mathrm{m/s}$

(B) $20\ \mathrm{m/s}$

(C) $40\ \mathrm{m/s}$

(D) $\frac{1}{40}\ \mathrm{m/s}$

Q.18

In a mass spectrometer used for measuring the masses of ions, the ions are initially accelerated by an electric potential V and then made to describe semicircular path of radius R using a magnetic field B. If V and B are kept constant, the ratio $\left(\frac{\text{charge on the ion}}{\text{mass of the ion}}\right)$ will be proportional to

(A) $1/\mathrm{R}^2$

@IITJEE_Advanced

(B) R^2

(C) R

(D) $1/R$

Q.19

The deflection in a galvanometer falls from 50 division to 20 when a 12 ohm shunt is applied. The galvanometer resistance
is

(A) 18 ohm

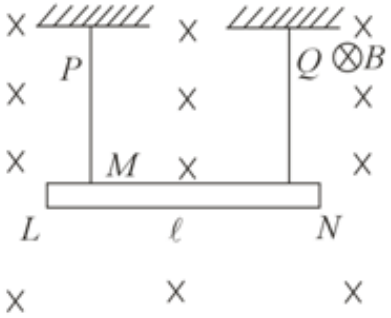
(B) $36ohm$

(C) $24ohm$

(D) $30ohm$

Q.20

A rod of mass M and length ℓ is suspended by two wires P and Q. A uniform magnetic field 'B' is directed into the page. When current through the rod is I. Then the tension in each supporting wire is-



(A) $\frac{mg}{2}$

(B) $2BI\ell$

(C) $Mg - BI\ell$

(D) $\frac{Mg-BI\ell}{2}$

Answers & Solutions

Q.1 Answer:

$B = \frac{\mu_0 I}{2\pi r}$ and $B = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$

Solution:

Figure shows the cross-sectional view of the cylinder.

Case (a) $r > R$. For this case, the Amperian loop is a circle of radius r concentric with the cross-section [Fig. 13.18

(a)]. For this loop, $L = 2\pi r$ and the current threading the loop is $i = I$. From Ampere's circuital

@IITJEE_Advanced

law.

$BL = \mu_0 i \Rightarrow B \times 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$

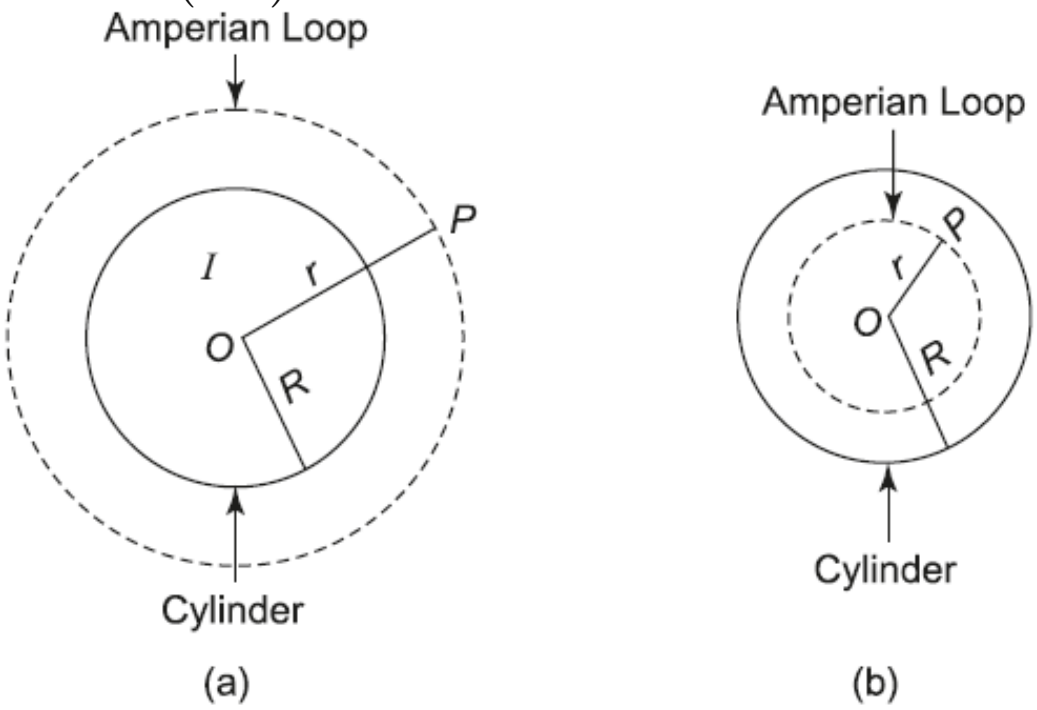
Case (b) $r < R$**. For this case,** $L = 2\pi r$ **[Fig. 13.18 (b)] and the current threading the loop is** $i =$ **current per unit cross-sectional area of the cylinder** $\times$ **cross-sectional area of the Amperian loop**

$= \frac{I}{\pi R^2} \times \pi r^2 = \frac{Ir^2}{R^2}$

From Ampere's law,

$B \times 2\pi r = \mu_0 i = \frac{\mu_0 I r^2}{R^2}$

$\Rightarrow B = \left(\frac{\mu_0 I}{2\pi R^2}\right) r$



Q.2 Answer:

$\frac{\mu_0 I}{\pi a}\left(1 - \frac{1}{2\sqrt{2}}\right)$

Solution:

Magnetic field at F **is**

$\mathrm{B} = \mathrm{B}_{AB} + \mathrm{B}_{BC} + \mathrm{B}_{CD} + \mathrm{B}_{DE} + \mathrm{B}_{EA}$

Since point F **lies in line with current elements** AB **and** $CD, B_{AB} = B_{CD} = 0$

Also $B_{DE} = B_{EA} = \frac{\mu_0 I}{4\pi a}(\sin 0^\circ + \sin 45^\circ) = \frac{\mu_0 I}{4\sqrt{2}\pi a}$

directed out of the page and towards the reader.

$B_{BC} = \frac{\mu_0 I}{4\pi BC/2}(\sin 45^\circ + \sin 45^\circ)$

directed into the page and away from the reader. Now

$BC = \sqrt{BF^2 + FC^2} = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{a}{\sqrt{2}}$

$\therefore B_{BC} = \frac{\mu_0 I}{4\pi\left(\frac{a}{2\sqrt{2}}\right)}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = \frac{\mu_0 I}{\pi a}$

directed into the page.

Now $B_{DE} + B_{EA} = \frac{\mu_0 I}{2\sqrt{2}\pi a}$ **directed out of the page.**

@IITJEE_Advanced

Since $B_{BC} > B_{DE} + B_{EA}$ the net field, B is directed into the page and has a magnitude

$B = \frac{\mu_0 I}{\pi a} - \frac{\mu_0 I}{2\sqrt{2}\pi a} = \frac{\mu_0 I}{\pi a}\left(1 - \frac{1}{2\sqrt{2}}\right)$

Q.3 Answer:

1.15

Solution:

For wire B bent into a circle,

$\mathrm{B}_1 = \left(\frac{\mu 0}{4\pi}\right)\left(\frac{2\pi \mathrm{I}}{\mathrm{R}}\right) = \left(\frac{\mu 0}{4\pi}\right)\left(\frac{2\pi \mathrm{I}}{\frac{\mathrm{L}}{2\pi}}\right)$

$= \left(\mu 0 \times \pi \times \frac{\mathrm{I}}{\mathrm{L}}\right) \ldots (1)$

For wire A bent into a square,

Refer to Fig. The magnetic field at O due to wire PQ is $(OT = a)$

$B_{PQ} = \frac{\mu_0 I}{4\pi a}(\sin 45^\circ + \sin 45^\circ)$

$= \frac{\mu_0 I}{2\sqrt{2}\pi a}$

$= \frac{4\mu_0 I}{\sqrt{2}\pi L} \quad \left(\because a = \frac{L}{8}\right)$

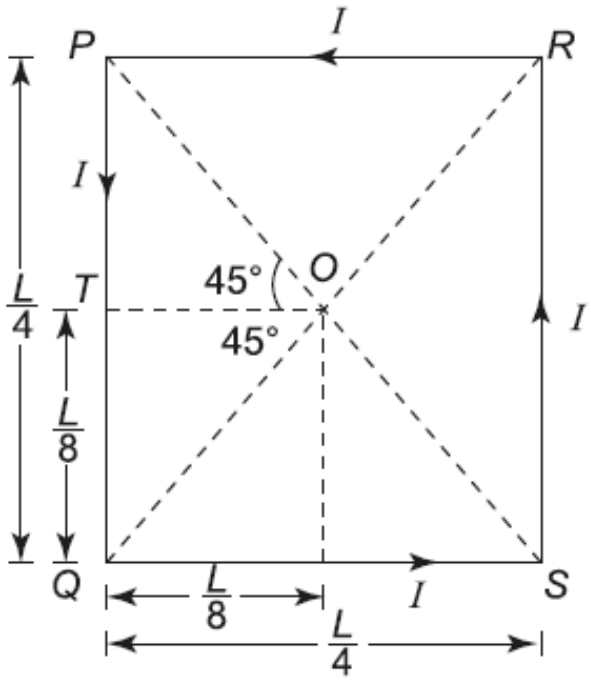
Since centre O of the square is at the same distance from each side of the square and each arm carries the same current, the magnetic field due to each side of the square is of the same and in the same direction. Hence the total magnetic field at O is

$B_2 = 4B_{PQ} = \frac{16\mu_0 I}{\sqrt{2}\pi L} = \frac{8\sqrt{2}\mu_0 I}{\pi L}$ **...(2)**

Dividing (2) by (1) we get

$\frac{B_2}{B_1} = \frac{8\sqrt{2}}{\pi^2} = \frac{8\times 1.41}{(3.14)^2} = 1.145 \approx 1.15$

Hence $B_2 > B_1$. The magnetic field at the centre due to the square loop will be greater than that due to the circular loop.



Q.4 Answer:

$\frac{\mu_0 \mathrm{qf}}{2\mathrm{R}}$

Solution:

@IITJEE_Advanced

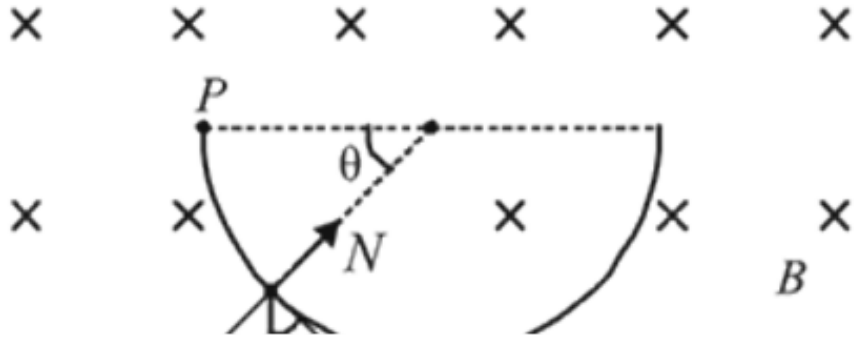
$i = \frac{q}{T} = qf$

$B = \frac{\mu_0 \hat{i}}{2R} = \frac{\mu_0 qf}{2R}$

Q.5 Answer:

$3mg\sin\theta + qB\sqrt{2gR\sin\theta}$

Solution:



$v = \sqrt{2gR\sin\theta}$

and $F = qvB$ Now $N - (F + \text{mgsin}\,\theta) = \frac{mv^2}{R}$

$\therefore \quad N = F + mg\sin\theta + \frac{mv^2}{R}$

$= qB\sqrt{2gR\sin\theta} + mg\sin\theta + 2mg\sin\theta$

$\therefore \quad N = 3mg\sin\theta + qB\sqrt{2gR\sin\theta}$

Q.6 Answer:

$\frac{\mu_0 I\theta}{4\pi r}$

Solution:

$\text{B} = \frac{\mu_0 \text{I}}{2\text{r}} \times \frac{\theta}{2\pi} = \frac{\mu_0 \text{I}\theta}{4\pi\text{r}}$

Q.7 Answer:

$250\mu\text{T}$

Solution:

$\text{B} = \frac{\mu_0 \text{ia}^2}{2(\text{x}^2+\text{a}^2)^{3/2}}$

$\text{B}' = \frac{\mu_0 \text{i}}{2\text{a}} = \frac{\mu_0 \text{ia}^2}{2\text{a}(\text{x}^2+\text{a}^2)^{3/2}} \left(\frac{(\text{x}^2+\text{a}^2)^{3/2}}{\text{a}^2}\right)$

$B' = \frac{B\cdot(x^2+a^2)^{3/2}}{a^3}$

Put $x = 4 \& a = 3 \Rightarrow B' = \frac{54(5^3)}{3\times3\times3} = 250\mu T$

@IITJEE_Advanced

Q.8 Answer:

$\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}$

Solution:



Let us compute the magnetic field due to any one segment :

$\mathrm{B} = \frac{\mu_0 \mathrm{i}}{4\pi(\mathrm{d}\sin\alpha)}(\cos 0^\circ + \cos(180-\alpha))$ **< smiles>C=CC < /smiles> Resultant field will be**

$\mathrm{B_{net}} = 2\,\mathrm{B} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}} \tan\frac{\alpha}{2} \Rightarrow \mathrm{K} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}$

Q.9 Answer:

$4B$

Solution:

Let I be current and l be the length of the wire. For Ist case : $B = \frac{\mu_0 I n}{2r} = \frac{\mu_0 I \times \pi}{l}$ **where** $2\pi r = l$ **and** $n = 1$ **For IInd Case :** $l = 2(2\pi r') \Rightarrow r' = \frac{l}{4\pi}$

$B' = \frac{\mu_0 n I}{2r'} = \frac{\mu_0 2I}{2\frac{l}{4\pi}} = \frac{4\mu_0 I\pi}{l} = 4B$

Q.10 Answer:

$\frac{\mu_0}{2\pi}$

Solution:

Since the fileds created by $\mathrm{i_1}$ **and** $\mathrm{i_2}$ **will be in opposite direction,**

$\mathrm{B} = \frac{\mu_0}{4\pi}\frac{2\mathrm{i_2}}{(\mathrm{r}/2)} - \frac{\mu_0}{4\mu}\frac{2\mathrm{i_1}}{(\mathrm{r}/2)} = \frac{\mu_0}{4\pi}\frac{4}{\mathrm{r}}(\mathrm{i_2} - \mathrm{i_1})$

$= \frac{\mu_0}{4\pi}\frac{4}{5}(5 - 2.5) = \frac{\mu_0}{2\pi}$

Q.11 Answer:

$1:3$

Solution:

Magnetic field due to the first loop, $\mathrm{B_1} = \frac{\mu_0 I}{2\mathrm{r}}$

For and wire, there are three loops $(\mathrm{n} = 3)$ **so total current enclosed by the loops**

$\mathrm{I_{en}} = \mathrm{I}/3 + \mathrm{I}/3 + \mathrm{I}/3 = \mathrm{I}$

@IITJEE_Advanced

Radius of second loop $= \mathrm{r}/3$ because the original wires are identical. magnetic field $\mathrm{B}_2 = \frac{\mu_0 \mathrm{I}_{\mathrm{cn}}}{2\mathrm{r}/3} = \frac{\mu_0 3\mathrm{I}}{2\mathrm{r}}$
$\frac{\mathrm{B}_1}{\mathrm{B}_2} = \frac{1}{3}$

Q.12 Answer:

$\sqrt{7}R$

Solution:

Magnetic field on axis of circular coil is $\frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}} = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}} = \frac{\mu_0 i_2 R^2}{2(R^2+x^2)^{3/2}}$
$x = \sqrt{7}R$

Q.13 Answer:

$10\mu\mathrm{T}$

Solution:

Current in the element $= J(2\pi r.dr)$ Current enclosed by ampere loop of radius $a/2$ $I = \int_0^{a/2} \frac{J_0 r}{a} \cdot 2\pi r \cdot dr$
$= \frac{2\pi J_0}{3a}\left(\frac{a}{2}\right)^3 = \frac{\pi J_0 a^3}{12}$
Applying ampere's law $B.2\pi \cdot \frac{a}{2} = \mu_0 \cdot \frac{\pi J_0 a^2}{12} \Rightarrow B = \frac{\mu_0 J_0 a}{12}$
On putting values, $B = 10\mu\mathrm{T}$

Q.14 Answer:

$\vec{B}_1 = -\vec{B}_2$

Solution:

Now it is given that a small current carrying element is placed at $(1,1,0)$ and carrying current in $+\mathrm{z}$ direction.
Then by using right hand thumb rule we can find the magnetic field lines which would be circular in this case having center at $(1,1,0)$
Now since origin and $(2,2,0)$ are two points which will lie on a circular centered at $(1,1,0)$ therefore, magnitude of magnetic field will be same at both the points.
And we find that origin and $(2,2,0)$ are diametrical opposite points therefore, direction of magnetic field at origin will be just opposite of the direction of magnetic field at $(2,2,0)$

@IITJEE_Advanced

Q.15 Answer:

10^{-7} T

Solution:

$0 = 0 + 1 \times 10 = 10 \text{rad/sec}^2 \quad \therefore v = r\omega = 1 \times 10 = 10 \text{ m/s}$

$\vec{B} = \frac{\mu_0(\vec{v} \times \vec{r})}{4\pi^3} \quad \Rightarrow |\vec{B}| = \frac{\mu_0 qv}{4\pi r^2}; B = \frac{10^{-7} \times 0.1 \times 10}{(1)^2} = 10^{-7} \text{ T}$

$\therefore$ **(b)**

Q.16 Answer:

2450Ω in series

Solution:

$R_g = 50\Omega, I_g = 25 \times 4 \times 10^{-A}\Omega = 10^{-2} \text{ A}$

Range of $V = 25$ **volts** $V = I_g\left(R_e + R_g\right)$

$\therefore R_e = \frac{V}{I_g} - R_g = 2450\Omega$

Q.17 Answer:

40 m/s

Solution:

As electron move with constant velocity without deflection. Hence, force due to magnetic field is equal and opposite to force due to electric field.

$\text{qvB} = \text{qE} \Rightarrow \text{v} = \frac{\text{E}}{\text{B}} = \frac{20}{0.5} = 40 \text{ m/s}$

Q.18 Answer:

$1/\text{R}^2$

Solution:

In mass spectrometer, when ions are accelerated through potential V $\frac{1}{2}mv^2 = qV$

As the magnetic field curves the path of the ions in a semicircular orbit

$Bqv = \frac{mv^2}{R} \Rightarrow v = \frac{BqR}{m}$

Substituting (ii) in (i) $\frac{1}{2}m\left[\frac{BqR}{m}\right]^2 = qV$

or $\frac{q}{m} = \frac{2V}{B^2R^2}$

Since V **and** B **are constants,** $\therefore \frac{q}{m} \propto \frac{1}{R^2}$

Q.19 Answer:

@IITJEE_Advanced

18 ohm

Solution:

$I = 50k; I_g = 20k$, **where** k **is the figure of merit of galvanometer;** $S = I_g R_g (I - I_g)$ **; so** $12 = \frac{20k \cdot R_g}{(50k - 20k)}$

On solving we get $R_g = 18 ohms.$

Q.20 Answer:

$\frac{Mg - BI\ell}{2}$

Solution:

$mg - BI\ell = 2T$

$T = \frac{mg - BIC}{2}$

@IITJEE_Advanced